

Article

One Model, Many Arms: Amortized Multimodal Inverse Kinematics Across a Continuous Family of Soft Continuum Manipulators

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Abstract: A learned inverse-kinematics (IK) model for a soft continuum arm is trained for one arm. Yet no two fabricated soft arms are the same arm: segment lengths, stiffness, and actuation gain vary by fabrication tolerance, and a model fitted to one instance is silently wrong on the next. We ask whether a single conditional diffusion model can amortize inverse kinematics across a continuous family of soft arms—solving IK for morphologies it never saw—and what makes that possible. On a piecewise-constant-curvature pneumatic arm whose four segment lengths and four curvature gains are each drawn per sample from a continuous box, a diffusion model conditioned on the morphology vector reaches 0.41 ± 0.04 mm best-of-32 tip error with 100% success on held-out in-distribution morphologies (three seeds)—within 0.06 mm of specialists trained on 5×10^5 samples of their own arm, with zero samples of any specific arm. A controlled ablation isolates the cause: the identical randomized data with the morphology *withheld* yields 12.7 ± 0.2 mm and 12% success, a $31\times$ gap. A direct test shows what the withheld-morphology model learned instead: every one of its samples is a correct solution for *some* arm in the family (100%), almost never for the queried one (0–2%)—it learned the marginal over arms, which also explains why gain randomization recovers nothing under dynamics-model change. Generalization is flat across the training box and degrades gracefully just beyond it—to 10–15% past the edge when extrapolation enlarges the workspace, 25–33% when it shrinks it—before a sharp cliff. Deterministic and mixture-density regressors trained on the same data do not amortize (14–16 mm in-distribution), because the morphology-conditioned inverse map is more multimodal, not less. The amortized model recovers 0.96 ± 0.01 of each unseen arm’s enumerated solution modes, matching a specialist’s recall at equal mode count. We report the single-arm advantage, its redundancy mechanism, and the limits of transfer under dynamics-model change, with three-seed statistics on every headline comparison. Code, data, and per-experiment result files are released.

Keywords: soft robotics; continuum manipulators; inverse kinematics; diffusion models; amortized inference; domain randomization; multimodal learning

1. Introduction

A multi-segment pneumatic continuum arm with three chambers per segment has more actuation degrees of freedom than task degrees of freedom: for a four-segment arm, a 12-D actuation vector determines a 3-D tip position. The inverse map is one-to-many, and its preimages are not a connected null space—distinct arm shapes reach the same tip. A deterministic learned IK regressor trained by motor babbling [1] fits the conditional *mean* of this multimodal distribution and can return an actuation that realizes none of the true solutions. A conditional diffusion model [2,3] learns the full conditional distribution $p(\mathbf{q} | \text{target})$ and can sample every solution mode. We establish below, with three-seed

statistics, that this advantage is real, that it is specifically a redundancy effect, and that it does not survive a change of dynamics model.

Every learned soft-arm IK method we are aware of, including the single-arm models we begin with, trains one model for one arm. This paper starts from the observation that *one arm* is a fiction. A soft manipulator is a fabricated object: its segment lengths depend on assembly, its bending stiffness on material batch and geometry (for a rod-backed arm stiffness goes as the fourth power of diameter, so a $\pm 4\%$ tolerance is a $\pm 17\%$ stiffness range), and its actuation gain on chamber or tendon geometry. Two arms built from the same drawing are two different arms. A model that must be retrained for each one—at 5×10^5 simulated samples per arm—is a model that does not scale past the bench it was trained on.

The question is therefore whether a single model can *amortize* inverse kinematics across a family of arms: trained once on a continuous distribution of morphologies, and asked to solve IK for a specific morphology it never saw, given only that morphology’s parameters as a conditioning input. This is the soft-robotic instance of a question now central to rigid-robot learning, where diffusion IK has been made to generalize across kinematic-tree *topologies* [4,5]. Soft arms pose it differently: the morphology is not a discrete topology but a continuous vector, and the conditional distribution the model must represent is more multimodal, not less—the same tip is reached by different shapes on different arms.

There is a specific reason to doubt the answer is yes. Domain randomization (DR) [6]—training on randomized arm parameters so that the model is robust to their variation—recovers *nothing* under model change in our experiments, while degrading in-domain accuracy fivefold (Sec. 7). If randomizing the morphology hurts, why would a randomized family be learnable at all? The answer this paper gives is that DR randomized the arm but *hid* it from the model. With the morphology withheld, the model learns the marginal over the family—every sample a valid solution for some arm, and for the queried arm only by chance; with it exposed, the model can learn the inverse map as a function of morphology. The difference is one conditioning vector, and it is the difference between a 12.7 mm model and a 0.41 mm one.

Our contributions:

1. **Amortization at specialist accuracy.** A single morphology-conditioned diffusion model reaches 0.41 ± 0.04 mm best-of-32 tip error and 100% success on held-out in-distribution morphologies, within 0.05 mm of a per-arm specialist that spent the same data budget on one arm alone (Sec. 5).
2. **The information, not the data.** The same randomized training set with the morphology withheld—domain randomization—reaches 12.7 ± 0.2 mm and 12% success. Withholding the morphology costs $31\times$ in accuracy and lands *below* the single-arm model even on that arm. A direct test shows the withheld-morphology model learned the marginal over arms: 100% of its samples solve the target on some arm in the family, 0–2% on the queried one (Sec. 5.2).
3. **A measured generalization envelope.** Accuracy is flat across the training box, degrades gracefully just past its edge, then cliffs; the margin is asymmetric—25–33% when extrapolation shrinks the workspace, 10–15% when it enlarges it (Sec. 5.3).
4. **Only a multimodal model amortizes.** Deterministic and mixture-density regressors trained on the identical morphology-conditioned data reach 14–16 mm in-distribution. Amortization raises the multimodality of the inverse map, and only a model that represents it can exploit the conditioning (Sec. 5.6).
5. **The single-arm advantage and its limits, with three-seed statistics.** Diffusion IK dominates MLP and MDN baselines on one arm, recovering 0.83 ± 0.03 of enumerated modes where both recover none; the advantage is isolated as redundancy resolution by two controlled manipulations; and the learned inverse map collapses under dynamics-model change on every seed, with the recovery mechanisms quantified (Secs. 4, 6, 7). The amortization result is read against this honest account of what a learned inverse map can and cannot do.

Negative and bounded results are reported with the same prominence as positive ones; Sec. 9 states the limitations explicitly.

2. Related Work

IK for continuum arms. Non-learned approaches solve continuum IK by optimization through a model at query time: Jacobian schemes for nonconstant-curvature arms [1], real-time FEM inversion [7], and metaheuristic search with obstacle constraints [8]. The constant-curvature kinematics underlying our simulator are reviewed in [9]; STIFF-FLOP-style pneumatic arms motivate the three-chamber actuation model [10]. Learned approaches fit a deterministic target-to-actuation map on motor-babbling data [1]; mixture density networks [11] are the classical answer to multimodal regression and serve as our learned multimodal baseline. Every learned soft-arm IK method we are aware of is trained for one arm.

Diffusion and flow models for robots. Denoising diffusion models [2] with DDIM sampling [3] and classifier-free guidance [12] underpin diffusion policies [13]. IKDiffuser [4] solves IK for rigid kinematic trees and generalizes across *topologies* by tokenizing end-effectors; generative graphical IK [5] likewise generalizes over rigid-robot graphs. Neither addresses compliant bodies, and both treat morphology as a discrete structure rather than a continuous parameter. For continuum robots, flow matching has recently been used to learn a *forward* shape model conditioned on actuation across multi-module tendon-driven arms [14]; our problem is the inverse, and the quantity we condition on is the morphology, not the actuation.

Domain randomization and amortized inference. DR [6,15] trains on randomized simulator parameters to induce robustness; the parameters are, by design, not exposed to the policy. The alternative—exposing the parameters as conditioning so that a single model serves a family—is amortized inference in the sense of [16]. In rigid-body control it is the universal policy of [17], which conditions on identified dynamics parameters, and the design principle behind morphology-conditioned locomotion [18]. We are not aware of it having been applied to soft-arm IK, nor of a controlled comparison in any setting that holds the training data fixed and varies only whether the parameters are exposed.

3. Setup

3.1. The arm family

The training simulator is a four-segment piecewise-constant-curvature (PCC) arm [9]. Segment i has length L_i and three 120° -spaced chambers with normalized pressures $\mathbf{q} \in [0, 1]^{12}$; a linear chamber-to-curvature map with gain g_i yields per-segment curvature components (k_x, k_y) , and forward kinematics composes constant-curvature arcs. The model is batched and differentiable in $\mathbf{q}$, and—new here—in the per-segment lengths and gains, so that one forward pass evaluates a batch of *different* arms.

A *morphology* is the vector $\boldsymbol{\theta} = (L_1, \dots, L_4, g_1, \dots, g_4) \in \mathbb{R}^8$. The *nominal* arm of Sec. 4 is $L_i = 60$ mm, $g_i = 25$ (total length 240 mm). The *training box* is $L_i \in [40, 80]$ mm, $g_i \in [15, 35]$, each drawn independently and uniformly per sample; the nominal arm lies inside it. Total arm length across the box spans 160–320 mm and maximum reach 120–353 mm among the test morphologies below.

3.2. Three models, one training set

The design turns on a single contrast, so it is stated precisely. One training set of 5×10^5 samples is generated: for each, a morphology $\boldsymbol{\theta}$ is drawn from the box, an actuation $\mathbf{q}$ is drawn uniformly, and the tip is computed on *that* arm. Three models are then trained.

Amortized conditioned on $[\text{tip}, \boldsymbol{\theta}]$ (11-D). The model is told which arm each sample came from.

Blind conditioned on $[\text{tip}]$ only, on the *identical* samples. The model sees the same diversity of arms and is not told which one any sample is. This is domain randomization, extended from the gain-only randomization of Sec. 7 to length and gain.

Nominal conditioned on [tip], trained on 5×10^5 samples of the nominal arm alone. This is the single-arm model of Sec. 4, and represents current practice.

Amortized and Blind are trained on the same samples with the same architecture, optimizer, and schedule; they differ only in whether the eight morphology dimensions are included in the condition. Any gap between them is attributable to that information, not to the data.

3.3. Models and protocol

The diffusion model is a residual MLP denoiser (six FiLM-conditioned blocks, width 512) trained as a DDPM ($T=1000$, cosine schedule) with classifier-free guidance (condition dropout 0.1); sampling uses DDIM with 20 steps and guidance weight 1.5; both were fixed by ablation on the nominal arm (guidance has an inverted-U optimum at 1.5; accuracy and recall saturate by ~ 20 steps). Baselines are an MLP regressor and an MDN (eight components) matched in width, each trained in all three regimes, plus a classical optimization-based reference (gradient-descent IK through the differentiable PCC model from 32 random restarts, Adam, 300 iterations; grad-IK). Unless stated otherwise we sample $K=32$ candidates per target and report best-of- K ; success means tip error under 5 mm.

Held-out morphologies. Test arms fall in three bands. *Nominal* is the single-arm model’s arm, inside the box, so an interpolation test. *In-distribution* is five morphologies drawn from the box with a separate seed; with eight continuous parameters, no test morphology coincides with any training sample. *Extrapolation* is five arms outside the box: all segments long ($L=90$ mm), short (30 mm), stiff ($g=40$), compliant ($g=10$), and a mixed case alternating long-stiff and short-compliant segments. Every metric is computed on the test arm’s own kinematics: a PCC arm is instantiated at that morphology and the full metric suite runs on it. Accuracy uses 512 targets per morphology sampled from that arm’s reachable set.

Ground-truth modes. For mode recall we enumerate the true IK solution set by a 2×10^6 -sample sweep on the test arm, keeping actuations within tolerance and clustering with DBSCAN [19] ($\epsilon=12$, $\text{min_samples}=5$; match radius 15) in the 8-D per-segment curvature space, which removes the chamber map’s per-segment null space. Evaluation targets on the nominal arm average 2.05 modes.

Statistics. Every $\pm$ is the population standard deviation over three training seeds. Where a result is single-seed it is marked so.

4. The Single-Arm Advantage

We first establish the single-arm result the amortization study builds on: on the nominal arm, with three training seeds, does a diffusion model beat deterministic and mixture-density regressors, and is the margin a multimodality margin?

Table 1. Single-arm comparison on the nominal arm (mean $\pm$ std, 3 seeds; $K=32$). Mode recall and diversity over successful samples only. grad-IK is one optimization run per target and needs the differentiable simulator at query time. Source: `multiseed_results.json` (aggregate); single-sample row from `results_seed{0,1,2}.json`.

	Diffusion	grad-IK	MLP	MDN
Tip err. best-of- K (mm)	0.44 ± 0.03	0.013	13.4 ± 0.0	17.3 ± 0.1
Tip err. single sample (mm)	1.69 ± 0.15	—	13.4 ± 0.0	83.6 ± 2.7
Success @ 5 mm	1.00	1.00	0.060	0.027
Mode recall	0.83 ± 0.03	0.85	0.00	0.00
Diversity (curv. space)	28.2 ± 0.2	29.0	0.0	0.0
Obstacle success	0.978 ± 0.002	0.980	0.040	0.018
Query time (ms/target)	38.6 ± 4.9	428	0.013	0.022

Best-of- K diffusion reaches 0.44 mm at 100% success; the MLP, structurally limited to the conditional mean, plateaus at 13.4 mm, and the MDN does not improve on it. Against enumerated

solution sets, diffusion recovers 0.83 ± 0.03 of modes and both learned baselines recover none—the MDN not through component collapse but because no component is accurate enough to count as a success. When the task requires any of K samples to be simultaneously accurate and collision-free among three random spherical obstacles, diffusion succeeds on 97.8% of trials against 4.0%. grad-IK, with the simulator in hand, is the accuracy ceiling at 0.013 mm and recovers modes at a rate on par with diffusion; the learned model’s distinction is that it answers from samples alone at a $5\text{--}11\times$ lower query cost. On every task metric the *worst* diffusion seed outperforms the *best* learned-baseline seed—for tip error, 0.49 vs. 13.33 mm, a gap exceeding $400\times$ the across-seed standard deviation. The MDN’s zero recall is not a training pathology: NLL decreases monotonically, 7.8 of 8 components remain effective, and sweeping the component count over $\{4, 8, 16, 32\}$ never surpasses the plain MLP. These conclusions are not artifacts of the enumeration hyperparameters: across a 3×3 grid of DBSCAN ϵ and match radius, the learned baselines stay at zero everywhere while diffusion and grad-IK trade the lead.

5. Amortization Across Morphologies

Table 2. The central comparison: best-of-32 tip error and success at 5 mm, mean over the morphologies in each band. Amortized and Blind were trained on the identical 5×10^5 samples; Nominal on 5×10^5 samples of the nominal arm. Mean $\pm$ std over three training seeds. Source: `results_morph_3seed.json`.

Band	Amortized		Blind		Nominal	
	err (mm)	succ.	err (mm)	succ.	err (mm)	succ.
Nominal arm	0.48 ± 0.06	100%	5.36 ± 0.21	$48 \pm 5\%$	0.43 ± 0.04	100%
In-distribution	0.41 ± 0.04	100%	12.68 ± 0.21	12%	32.8 ± 0.2	5%
Extrapolation	6.19 ± 0.31	$62 \pm 3\%$	34.7 ± 0.1	2%	61.1 ± 0.7	0.4%

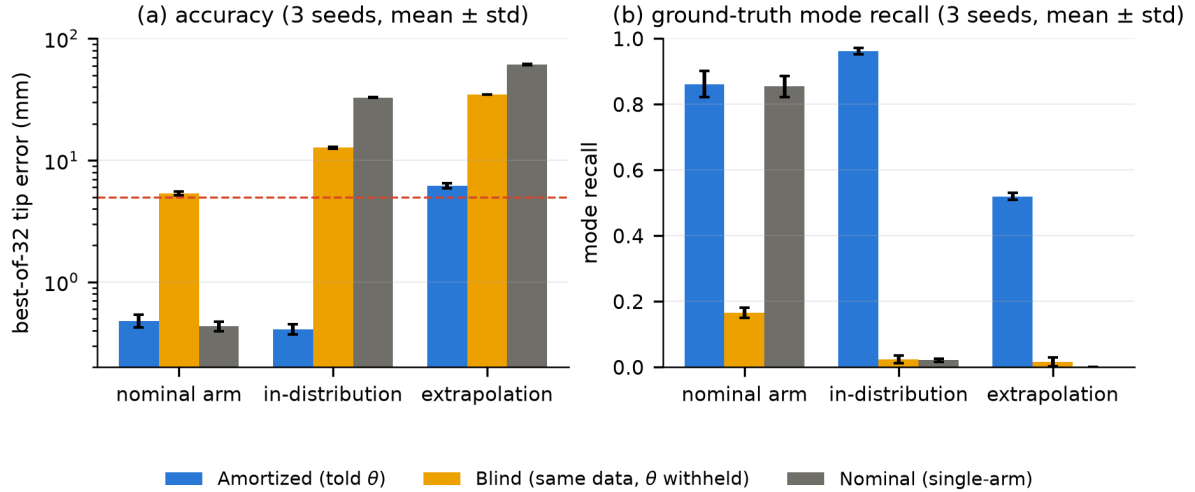


Figure 1. The central comparison. (a) Best-of-32 tip error (log scale; dashed line is the 5 mm success tolerance) and (b) ground-truth mode recall, by test band, for the three models of Sec. 3.2. Amortized and Blind share identical training data. Mean $\pm$ std over three training seeds.

5.1. Amortization at the specialist’s accuracy

Table 2 and Figure 1 give the paper’s central result. On the five held-out in-distribution morphologies, the amortized model reaches 0.41 ± 0.04 mm best-of-32 tip error with 100% success on every arm and every seed (per-seed means 0.40, 0.46, 0.37 mm; per-arm three-seed means 0.31–0.56 mm). On the nominal arm—which it never saw as an exact morphology—it reaches 0.48 ± 0.06 mm, against

0.43 $\pm$ 0.04 mm for the specialist that spent its entire 5×10^5 -sample budget on that arm alone. The cost of generalizing across the family, measured on the family’s own nominal member, is 0.05 mm—within one standard deviation, and 1% of the success tolerance.

A reviewer will ask whether that holds off the nominal arm, so we trained a second specialist, on one of the held-out in-distribution morphologies (single seed; 5×10^5 samples of that arm alone). It reaches 0.33 mm; the amortized model on the same arm reaches 0.39 $\pm$ 0.03 mm (three seeds), with identical mode recall (1.00) and diversity (21.1 vs. 21.1). The cost of amortization is 0.06 mm here and 0.05 mm on the nominal arm: consistent, and a tenth of a percent of arm length. Source: `results_specialist_indist0.json`.

The specialist, conversely, is useless off its arm: 32.8 $\pm$ 0.2 mm and 5% success in-distribution, 61.1 $\pm$ 0.7 mm and 0.4% extrapolating. Its errors on individual in-distribution arms range from 10 to 65 mm; on the long-segment arm, 115 mm. A model trained on one arm has learned a lookup table for that arm, and the table is wrong everywhere else.

5.2. The information, not the data

The Blind model was trained on the identical samples as the Amortized model. It reaches 12.68 $\pm$ 0.21 mm and 12% success in-distribution—a $31 \times$ gap in error, stable to ± 0.2 mm across seeds, attributable to nothing but the eight conditioning dimensions. On the nominal arm it reaches 5.36 $\pm$ 0.21 mm and 48 $\pm$ 5%: training on a randomized family *without* the morphology lands worse than the single-arm specialist on that very arm.

Table 3. What the Blind model learned. For each sample $\mathbf{q}$ drawn for target $\mathbf{t}$ on the queried arm: is it a valid solution (< 5 mm) on the queried arm, and is it valid on *some* arm in the training box (random search over 2×10^4 morphologies; a lower bound, since the search rarely lands on any specific arm exactly)? 20 targets, 32 samples each; random actuations are the control. Seed 0. Source: `results_blind_marginal.json`.

Queried arm	Samples from	valid on queried arm	valid on some arm
nominal	Blind	2.2%	100.0%
	Amortized	99.8%	100.0%
	random $\mathbf{q}$	0.0%	7.7%
in-dist	Blind	0.0%	100.0%
	Amortized	99.7%	96.1%
	random $\mathbf{q}$	0.0%	6.6%
extrap. stiff	Blind	0.0%	99.7%
	Amortized	37.7%	40.0%
	random $\mathbf{q}$	0.0%	4.2%

What did the Blind model learn instead? The hypothesis is that with the morphology withheld, the model fits the marginal $p(\mathbf{q} | \mathbf{t}) = \int p(\mathbf{q} | \mathbf{t}, \theta) p(\theta) d\theta$: every sample is a correct actuation for *some* arm in the family, and for the queried arm only by chance. This is testable, and Table 3 tests it. On every queried arm, essentially every Blind sample (99.7–100%) is a valid solution for some morphology in the training box, while only 0–2% are valid for the arm actually queried. Random actuations pass the same search 4–8% of the time, so the test discriminates. The Blind model is not confused or under-trained; it is a good sampler from the wrong distribution.

The prediction is quantitative. If each Blind sample is independently valid on the nominal arm with probability 0.022, best-of-32 success should be $1 - (1 - 0.022)^{32} = 51\%$; the measured rate is 48 $\pm$ 5% (Table 2). Blind’s nonzero success is the chance that one of 32 draws from the marginal happened to land on the queried arm.

This is the mechanism behind domain randomization’s failure on this problem. The finding that gain-DR degrades in-domain accuracy from 0.44 to 2.18 mm and recovers nothing under model

change (Sec. 7) is the same effect, observed without the test that explains it. DR does not fail because randomization is the wrong idea; it fails because withholding the parameters turns a conditional into a marginal, and a marginal over arms is not a solver for any of them.

The table's Amortized rows carry a second finding. In-distribution, an amortized sample valid on the queried arm is also valid on some arm (96%, the search's lower bound). On the extrapolated stiff arm, only 40% of amortized samples are valid *anywhere* in the box: when the model fails past its training family, its output is not a solution for a neighbouring arm but a solution for none. Extrapolation failure is breakdown, not confusion.

5.3. The generalization envelope

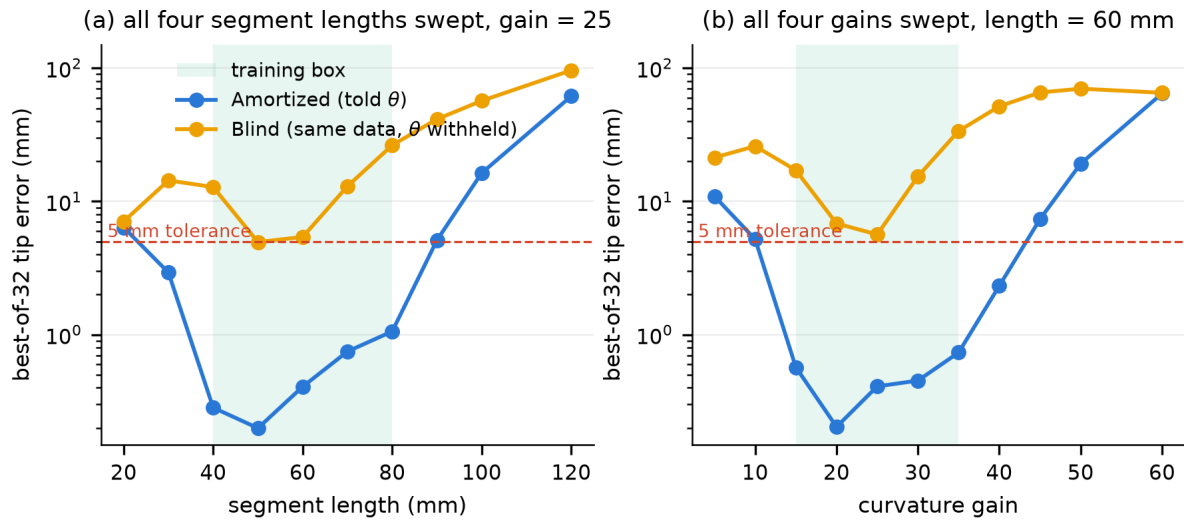


Figure 2. Generalization envelope. One morphology axis is swept with the other held at nominal, all four segments together; shaded region is the training box. The amortized model is flat inside the box, degrades gracefully just past its edge, then cliffs. The Blind model is best at the box centre and worse toward every edge—the signature of a model that learned the marginal over arms (Table 3). Seed 0. Source: `results_envelope_seed0.json`.

Table 4. Generalization envelope: one morphology axis swept with the other at nominal, all four segments together. Best-of-32 error and success. Shaded rows are inside the training box. Seed 0; source: `results_envelope_seed0.json`.

L	Segment length (mm), $g=25$		g	Curvature gain, $L=60$	
	Amortized	Blind		Amortized	Blind
20	6.4 / 31%	7.1 / 44%	5	10.9 / 20%	21.3 / 13%
30	2.9 / 94%	14.4 / 1%	10	5.2 / 56%	26.0 / 12%
40	0.28 / 100%	12.8 / 7%	15	0.57 / 100%	17.1 / 14%
50	0.20 / 100%	5.0 / 57%	20	0.20 / 100%	6.8 / 36%
60	0.41 / 100%	5.4 / 50%	25	0.41 / 100%	5.6 / 41%
70	0.75 / 100%	13.0 / 12%	30	0.45 / 100%	15.4 / 6%
80	1.06 / 99.6%	26.4 / 1%	35	0.73 / 99.6%	33.8 / 1%
90	5.1 / 60%	41.4 / 0.4%	40	2.3 / 98%	51.7 / 0%
100	16.3 / 3%	56.8 / 0%	45	7.4 / 30%	65.6 / 0%
120	61.6 / 0%	96.2 / 0%	50	19.2 / 5%	70.1 / 0%
			60	64.5 / 0%	65.3 / 0%

A single extrapolation number would mislead; what a reader needs is the shape of the degradation. Figure 2 and Table 4 sweep each axis from well inside the box to well outside. Inside, the amortized

model is flat: 0.20–1.06 mm and $\geq 99.6\%$ success across the entire range of both axes. Just past the edge it degrades gracefully— $L=90$ mm (12% past) gives 5.1 mm at 60%; $g=40$ (14% past) gives 2.3 mm at 98%—and then cliffs: $L=120$ mm and $g=60$ are 62 and 65 mm at 0%.

The envelope is asymmetric. Shortening extrapolates further than lengthening ($L=30$ mm, 25% below the box: 2.9 mm at 94%; $L=90$ mm, 12% above: 5.1 mm at 60%), and stiffening further than softening ($g=40$: 98%; $g=10$, 33% below: 56%). Both directions that extrapolate well shrink the workspace; both that extrapolate poorly expand it into regions the training arms never reached. The model generalizes over the morphology axis but not to tip positions outside the union of training workspaces.

The Blind column is the marginal made visible: best at the box centre ($L=50$ – 60 mm, $g=20$ – 25), where a draw from the marginal is likeliest to be a draw for a nearby arm, worse toward every edge, and never above 57% success anywhere.

5.4. Data efficiency

A single-arm specialist’s accuracy as a function of its training-set size is shown in Figure 3: 12.0 mm at 5×10^3 samples, 1.86 at 2×10^4 , 0.78 at 5×10^4 , 0.51 at 2×10^5 , and 0.40 mm at 5×10^5 . Reaching sub-millimetre accuracy on one arm costs that arm $\sim 5 \times 10^4$ samples; reaching 0.4 mm costs it 5×10^5 .

The amortized model spent 5×10^5 samples across a continuous 8-D family—in expectation, zero samples at any specific morphology—and reaches 0.41 ± 0.04 mm on morphologies it never saw. Per arm served, its data cost is not a number that can be written down: the family is a continuum. What can be said is that one specialist’s total budget buys specialist-level accuracy on every arm in the box.

Amortization is not free at every budget. Trained on 10^5 and 2×10^5 total samples (single seed), the amortized model reaches 1.36 and 1.22 mm on held-out in-distribution arms, against 0.71 and 0.51 mm for a specialist given the same total on its own arm—roughly $2 \times$ worse, while still at $\geq 99\%$ success. At 5×10^5 the two converge (0.39 vs. 0.40 mm). Spreading a small budget over a family costs accuracy; spreading a large one does not. The crossover lies between 2×10^5 and 5×10^5 total samples for this family.

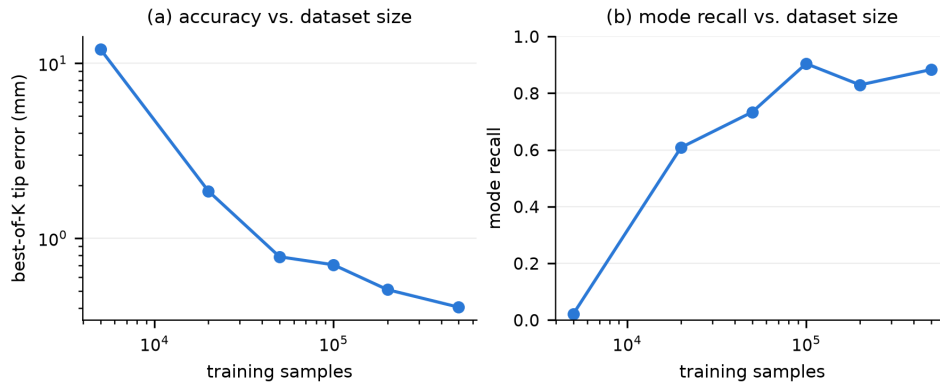


Figure 3. Single-arm dataset-size ablation on the nominal arm (single seed). (a) Best-of- K tip error and (b) mode recall vs. training-set size; both saturate by ~ 0.5 – 1×10^5 samples. The amortized model reaches the 5×10^5 -sample specialist’s accuracy on every in-distribution arm with the same total budget spread across the family.

5.5. Multimodality under amortization

Table 5. Mode recall and solution diversity on held-out morphologies, against ground-truth solution sets enumerated on each test arm (2×10^6 sweep, DBSCAN in curvature space; 20 targets per arm, successful samples only). Nominal band is one arm; in-distribution is five held-out arms; extrapolation is the three of five arms on which enumeration found modes (see text). Enumerated modes per target: nominal 2.05; in-distribution 1.00–1.90. Mean $\pm$ std over three training seeds. Source: `results_morph_3seed.json`.

Band	Amortized		Blind		Nominal	
	recall	div.	recall	div.	recall	div.
Nominal arm	0.86 ± 0.04	27.4 ± 0.5	0.17 ± 0.02	5.0 ± 1.6	0.85 ± 0.03	28.1 ± 0.2
In-distribution	0.96 ± 0.01	25.0 ± 0.2	0.02 ± 0.01	0.2 ± 0.3	0.02 ± 0.01	0.2 ± 0.2
Extrapolation	0.52 ± 0.01	6.4 ± 0.6	0.02 ± 0.01	0.0	0.00	0.0

The single-arm advantage was a multimodality advantage (Table 1). Whether amortization preserves it—whether the model still recovers the distinct solution modes of an arm it never saw—decides whether the amortized model is an IK *solver* or merely an IK *regressor* that happens to be conditioned well. Table 5 answers: it is a solver. On held-out in-distribution arms the amortized model recovers 0.96 ± 0.01 of enumerated modes with sampled diversity 25.0 ± 0.2 in curvature space. The Blind model recovers essentially none (0.02 ± 0.01): a sampler from the marginal does not merely miss the target, it misses every mode of every arm.

The in-distribution recall should not be read as exceeding the specialist’s on its own arm (0.85 ± 0.03). The five in-distribution arms average fewer enumerated modes per target (1.00–1.90; mean 1.35) than the nominal arm (2.05), and recall falls with mode count across them (from 1.00 on the one-mode arm to 0.88 on the 1.9-mode arm). On the nominal arm itself, where the mode count is matched, the amortized and specialist models are indistinguishable (0.86 ± 0.04 vs. 0.85 ± 0.03). The fair statement is that amortization preserves mode recall at the specialist’s level; it does not raise it.

In extrapolation, recall tracks success arm by arm: 0.88 recall at 92% success on the short arm, 0.58 at 55% on the compliant arm, 0.09 at 8% on the mixed arm. Past the training box the model does not find one solution while missing the others; when it reaches a target it recovers that target’s modes, and when it fails it fails entirely—the breakdown of Table 3, seen from the mode side. The diversity figure for this band (6.4 ± 0.6) is low chiefly because diversity is computed over successful samples and the mixed arm contributes almost none. (Two of the five extrapolation arms—all-long and all-stiff—are excluded from the recall average because the enumeration itself found no modes on them: the DBSCAN scale was calibrated to the nominal arm’s curvature range, and those two arms move furthest from it. The model reaches 97% success on the stiff arm, so this is an enumeration limit, not a model failure; see Sec. 9.)

5.6. Only a multimodal model amortizes

Table 6. Architecture comparison under amortization: all three models conditioned on $[\text{tip}, \theta]$ and trained on the same data. Best-of-32 error and success, seed 0. Source: `results_morph_seed0.json`.

Band	Diffusion	MLP	MDN
Nominal arm	0.42 / 100%	13.8 / 1%	18.1 / 2%
In-distribution	0.39 / 100%	14.1 / 3%	16.3 / 5%
Extrapolation	6.2 / 61%	31.8 / 15%	20.8 / 18%

Given the morphology, does any architecture amortize? Table 6 says no. The MLP and MDN, trained on the identical morphology-conditioned data, reach 14.1 and 16.3 mm in-distribution—no

better than their single-arm performance in Table 1, and no better than they manage without the morphology at all. The conditioning is not the bottleneck for them; their inability to represent a multimodal conditional is. Amortizing over morphology does not reduce the multimodality of the inverse map—it adds a dimension along which the solution set varies—so the models that could not represent it on one arm cannot represent it on a family. Diffusion’s advantage under amortization is not a capacity advantage: the MLP and MDN are matched in width, and the MLP’s validation loss on its own objective decreases monotonically to convergence. It fits the conditional mean as well as it can; the mean is not a solution.

6. The Advantage Is Redundancy, Specifically

If diffusion’s advantage—on one arm or on a family—is redundancy resolution, it should shrink when redundancy is removed and grow when it is added. Both predictions hold on one arm (Figure 4), and the amortization results extend the second.

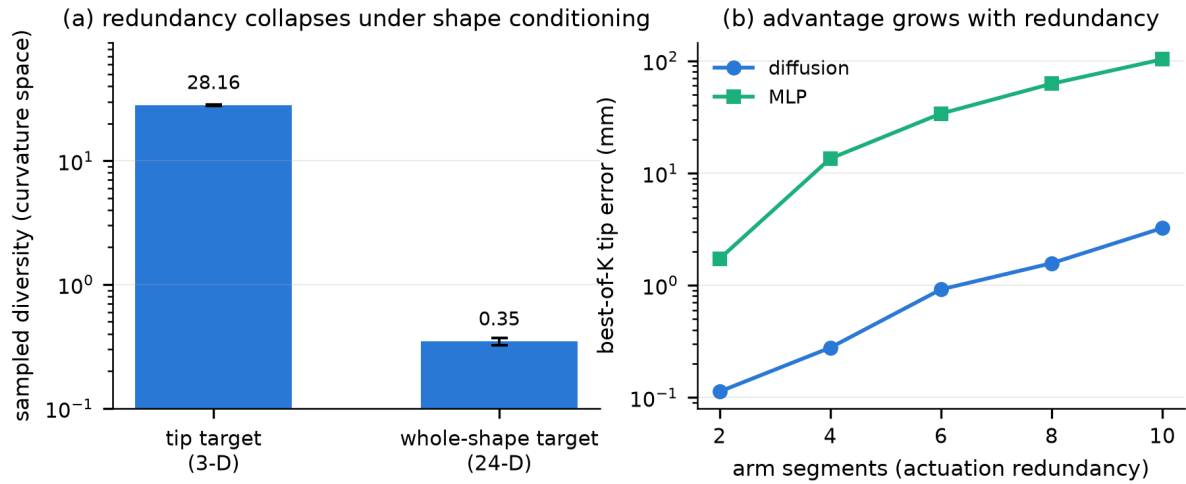


Figure 4. The diffusion advantage tracks actuation redundancy. (a) Replacing the 3-D tip target with a 24-D whole-shape target removes most of the redundancy: sampled diversity collapses $\sim 80\times$ (28.16 ± 0.22 to 0.346 ± 0.022 ; 3 seeds). (b) Increasing the number of segments (single seed per morphology) widens the diffusion–MLP accuracy gap monotonically from two to ten segments; from eight segments on, the MLP’s obstacle-task success is 0% while diffusion’s stays $\geq 99.5\%$.

Removing redundancy. Conditioning on the full 24-D body shape instead of the 3-D tip pins down almost the whole configuration. Diffusion’s sampled diversity collapses from 28.16 ± 0.22 to 0.346 ± 0.022 ($\sim 80\times$), and the MLP’s accuracy disadvantage shrinks from $\sim 30\times$ to $\sim 3\times$ (0.49 vs. 0.16 mm per-point error, both at 100% success, 3 seeds). When there is one answer, a deterministic regressor is nearly as good.

Adding redundancy. Across two-, six-, eight-, and ten-segment arms (single seed each), the MLP’s error grows monotonically from 1.7 to 103.2 mm with 0% success from eight segments on, while diffusion degrades gracefully (0.11 to 3.2 mm, $\geq 99.8\%$ success). Amortizing over morphology is a third way of adding redundancy—the same tip is now reachable by different shapes on different arms—and Sec. 5.6 shows the same ordering: the deterministic models fail, the multimodal one does not.

7. Under Dynamics-Model Change

Amortization across morphology is generalization across *parameters of the same model*. It is a different question whether a learned inverse map survives a change of *dynamics model*—executing PCC-trained actuations on a Cosserat-rod simulation of the same arm (PyElastica, 48 elements, 1.5 s damped horizon; [20,21]). It does not: accuracy collapses from 0.67 ± 0.03 to 139.9 ± 1.8 mm with

0% success on every seed ($n=20$ held-out targets, $K=8$; the PCC-domain figure differs from Table 1's 0.44 mm only by this smaller K), and the discrepancy is directional—no couple gain or damping constant aligns the two workspaces.

We characterized what recovers this loss and what does not; Table 7 summarizes and Figures 5 and 6 give the two central curves. Selection under an Elastica query budget improves every target but never solves one; a diversity-aware query ordering significantly beats a confidence ordering (paired $t=3.34$, $p\approx 0.002$); guided sampling is suggestive at best; fine-tuning on 10^4 target-domain samples yields the only nonzero success (67.1 ± 0.9 mm, 10%) at the price of catastrophic forgetting, and mixing source data back in collapses the gain below the first grid point—a retention-transfer cliff with no sweet spot.

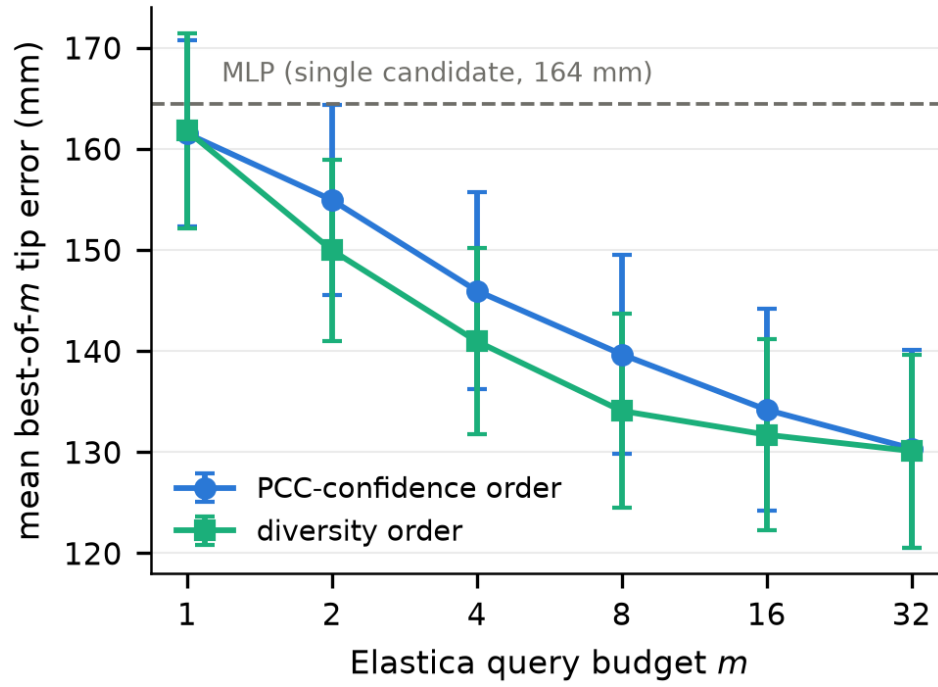


Figure 5. Candidate selection under an Elastica query budget ($n=45$ targets, $K=32$ candidates, mean $\pm$ SEM). A diversity-aware (farthest-point) ordering of which candidates to evaluate beats a confidence ordering; in the paired design the difference is significant at $m=8$ and $m=16$ ($t=3.34$ and 3.21 , $p\approx 0.002$).

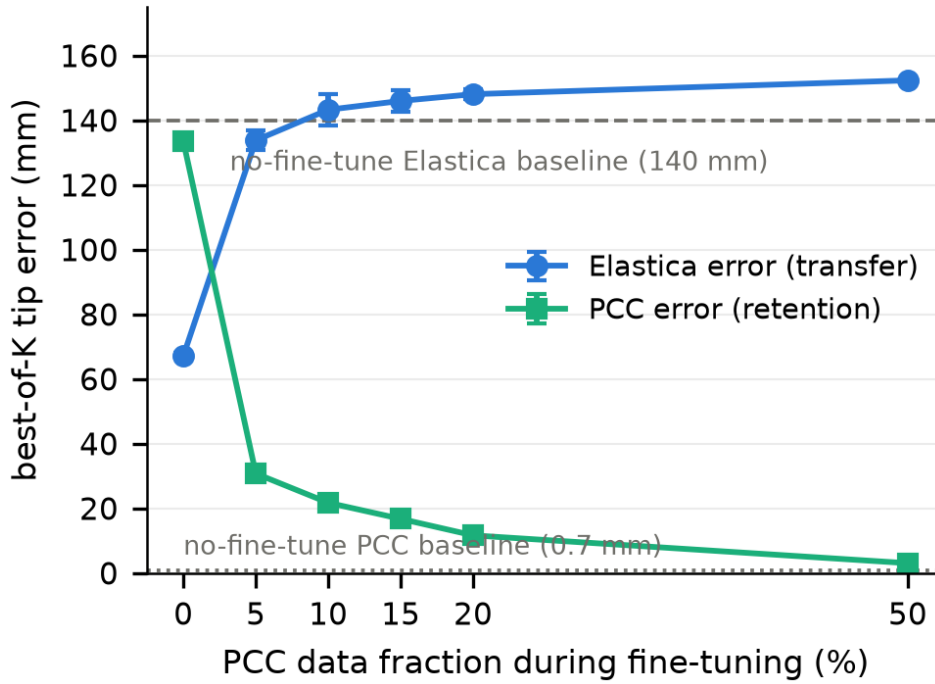


Figure 6. The retention–transfer cliff (mean $\pm$ std, 3 seeds; six mixing ratios). Fine-tuning on 10^4 Elastica samples: retention improves smoothly as source data is mixed back in, but the transfer gain collapses almost entirely by the first grid point (5%). There is no sweet spot.

Table 7. Recovery mechanisms under dynamics-model change (Elastica-domain performance; best configuration of each; each row on its own held-out target set, 15–45 targets).

Mechanism	err (mm)	succ.	Cost / caveat
none (top-1 candidate)	161.5	0%	no target queries
best-of-32 selection	130.4	0%	32 queries/target
+ diversity order	-6.7 at $m=8$	0%	same queries; $p \approx 0.002$
guided sampling	129.0	0%	400 labels; suggestive
gain randomization (blind)	140.6	0%	degrades source; no gain
fine-tune, 0% mix	67.1 ± 0.9	10%	forgets source (133 mm)
fine-tune, $\geq 5\%$ mix	≥ 133.8	$\leq 10\%$	retains source, forfeits gain

The row labelled *gain randomization* is the one this paper reinterprets. It was reported as a null result: randomizing the gain buys nothing. Sec. 5.2 shows why. That model was Blind; it had learned the marginal over gains, and a sample from that marginal is no likelier to suit the Cosserat arm than one from the nominal model. Whether an *Amortized* model—one that has learned the inverse map as a function of morphology—can be steered toward the Cosserat arm by inferring an effective morphology for it is the natural next question, and one this paper’s machinery makes askable. It is not answered here.

8. Discussion

What amortization buys. A practitioner with a fabricated soft arm whose segment lengths and gains can be measured—or estimated from a handful of calibration poses—can use one pretrained model rather than train one. The accuracy cost is under a tenth of a millimetre inside the training family and modest for the first 10–15% beyond it. Because the conditioning is a continuous vector, the model would also support a mode of use no per-arm model can: sweeping θ at a fixed target to ask

which morphology in the family reaches it most easily. We note the possibility; this paper does not evaluate it.

Why the ablation matters. That conditioning on identified parameters beats marginalizing over them is the premise of universal policies [17]. What the Blind result adds is a controlled measurement of the cost of marginalizing, on a problem where the conditional is multimodal: identical data, one conditioning vector removed, a $31\times$ accuracy gap, and a direct test (Table 3) showing the marginal was learned faithfully. The practical consequence is direct: when the randomized parameters can be observed at deployment, they should be conditioned on, not marginalized over. DR is the right tool only when the parameters are genuinely unobservable, and its cost in that case—a sampler whose every draw is right for some arm and wrong for yours—should be budgeted for.

Why only diffusion. The MLP’s converged fit alongside its 14 mm error is the cleanest statement of the paper’s mechanism. A model that fits the conditional mean can fit it arbitrarily well and still return an actuation that is not a solution, because the solutions are a set and the mean of a set is not in it. Amortization makes the set larger.

Relation to rigid-robot amortization. IKDiffuser [4] amortizes over discrete kinematic-tree topologies by tokenization. The soft case differs in that the morphology is a continuous parameter of a single topology, which makes interpolation and extrapolation meaningful notions—and we are not aware of a comparable envelope measurement for a generative IK model on any robot.

9. Limitations

(1) **Simulation only.** Every result is on a simulated arm; no claim is made about hardware. The motivating fabrication-variability argument is a reason to want amortization, not evidence that this model delivers it on a physical arm. (2) **The morphology must be known.** The amortized model is conditioned on θ ; it does not infer it. On a physical arm the eight parameters would have to be measured or calibrated, and the model’s sensitivity to error in them is not characterized here. (3) **One segment count.** The family varies lengths and gains at fixed $N=4$; segment count changes the actuation dimension and is outside this study’s conditioning scheme. (4) **Statistical coverage.** The single-arm comparison, shape-conditioning, transfer collapse, fine-tuning grid, the central amortization comparison (Table 2), and held-out mode recall (Table 5) carry three-seed statistics. The query-budget, ordering, guidance, hyperparameter, segment-count, envelope-sweep, architecture-comparison, amortized-scale, marginal-test, and second-specialist results are single-seed. (5) **Extrapolation is bounded.** The model does not generalize to tip positions outside the union of training workspaces; the cliff in Table 4 is a real boundary, not a tail. (6) **Dynamics-model change is not solved.** Sec. 7 is an honest boundary on what any learned inverse map in this family can currently do. (7) **Mode enumeration does not reach every arm.** The DBSCAN parameters were calibrated on the nominal arm; on the two extrapolation arms whose curvature-space scale is furthest from it, enumeration found no ground-truth modes and recall is undefined there. (8) **The test suite is eleven arms.** Five in-distribution draws and five extrapolation cases along chosen axes; the envelope sweeps one axis at a time with the other at nominal. Joint extrapolation on both axes is represented by a single mixed case. (9) **The marginal test is a lower bound.** Random search over 2×10^4 morphologies rarely lands on any specific arm, so “valid on some arm” understates; the Blind rows are at 100% regardless.

10. Conclusions

One conditional diffusion model, told which arm it is solving for, solves inverse kinematics for an entire continuous family of soft continuum arms—within a tenth of a millimetre of a per-arm specialist, with no data from any specific arm, and with graceful degradation for a measured distance beyond the family it was trained on. The same data with the morphology withheld does not: it learns the marginal over the family, every sample a solution for some arm and almost none for the one queried—measured directly, and predicting the observed success rate. Domain randomization’s failure on this problem is an information failure, and the ablation that shows it changes nothing but eight

conditioning dimensions. Only a model that represents the multimodal solution set can exploit that information, because under amortization the solution set varies with morphology as well as with target. For learned soft-arm IK, the path past the single-arm model is not more data from that arm but a model conditioned on which arm it is.

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Conflicts of Interest: The author declares no conflict of interest.

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